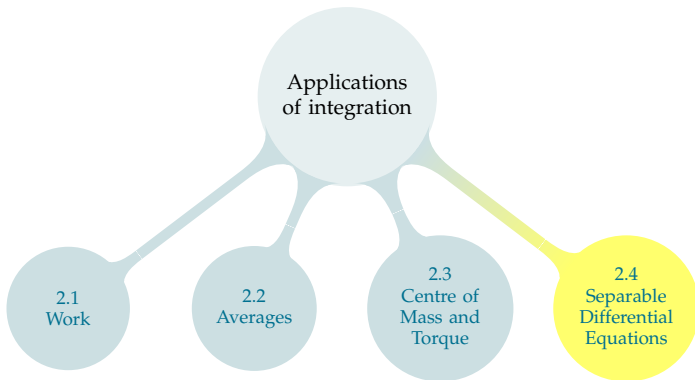


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Differential Equation

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Differential Equation

A **differential equation** is an equation for an unknown function that involves the derivative of the unknown function.

Differential equations play a central role in modelling a huge number of different phenomena. Here is a table giving a bunch of named differential equations and what they are used for. It is far from complete.

Newton's Law of Motion	describes motion of particles
Maxwell's equations	describes electromagnetic radiation
Navier-Stokes equations	describes fluid motion
Heat equation	describes heat flow
Wave equation	describes wave motion
Schrödinger equation	describes atoms, molecules and crystals
Stress-strain equations	describes elastic materials
Black-Scholes models	used for pricing financial options
Predator-prey equations	describes ecosystem populations
Einstein's equations	connects gravity and geometry
Ludwig-Jones-Holling's equation	models spruce budworm/Balsam fir ecosystem
Zeeman's model	models heart beats and nerve impulses
Sherman-Rinzel-Keizer model	for electrical activity in Pancreatic β -cells
Hodgkin-Huxley equations	models nerve action potentials

Disclaimer:

We are dipping our toes into a vast topic. Most universities offer half a dozen different undergraduate courses on various aspects of differential equations. We will just look at one special, but important, type of equation.

- We will first learn to **verify** solutions without **finding** them. (If you learned about differential equations last semester, this will be review.)

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We are dipping our toes into a vast topic. Most universities offer half a dozen different undergraduate courses on various aspects of differential equations. We will just look at one special, but important, type of equation.

- ▶ We will first learn to **verify** solutions without **finding** them. (If you learned about differential equations last semester, this will be review.)
- ▶ **Then**, we will learn to solve one particular type of differential equation.

DIFFERENTIAL EQUATIONS

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A **differential equation** is an equation involving the derivative of an unknown function.

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Definition

If a **function** makes a differential equation true, we say it **satisfies** the differential equation, or is a solution to the differential equation.

Example: $y = x^2$ and $y = x^2 + 1$ both satisfy the first differential equation

VERIFYING SOLUTIONS

Consider the equation

$$x + 2 = x^3 - x^2$$

How would you verify whether $x = 1$ satisfies the equation?

How would you verify whether $x = 2$ satisfies the equation?

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Plug x into the equation, check whether the left-hand side and the right-hand side are the same **number**.

VERIFYING SOLUTIONS

Consider the differential equation

$$\frac{dy}{dx} = 2y + 4x$$

How would you verify whether $y = e^{2x} - 2x$ satisfies the equation?

How would you verify whether $y = e^{2x} - 2x - 1$ satisfies the equation?

VERIFYING SOLUTIONS

Consider the differential equation

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How would you verify whether $y = e^{2x} - 2x - 1$ satisfies the equation?

Replace y and $\frac{dy}{dx}$ in the equation, check whether the left-hand side and the right-hand side are the same **function**.

VERIFYING SOLUTIONS

Consider the differential equation

$$\frac{dy}{dx} = 2y + 4x$$

Replace y and $\frac{dy}{dx}$ in the equation, check whether the left-hand side and the right-hand side are the same **function**.

- If $y = e^{2x} - 2x$, then $\frac{dy}{dx} = 2e^{2x} - 2$. Plug these into both sides of the differential equation, replacing anything depending on y :

$$\frac{dy}{dx} = 2y + 4x$$

$$2e^{2x} - 2 \stackrel{?}{=} 2(e^{2x} - 2x) + 4x$$

$$2e^{2x} - 2 \stackrel{?}{=} 2e^{2x}$$

Since the functions on the left and right are not the same function, $y = e^{2x} - 2x$ is **not** a solution to the differential equation.

VERIFYING SOLUTIONS

Consider the differential equation

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Replace y and $\frac{dy}{dx}$ in the equation, check whether the left-hand side and the right-hand side are the same **function**.

- If $y = e^{2x} - 2x - 1$, then $\frac{dy}{dx} = 2e^{2x} - 2$. Plug these into both sides of the differential equation, replacing anything depending on y :

$$\frac{dy}{dx} = 2y + 4x$$

$$2e^{2x} - 2 \stackrel{?}{=} 2(e^{2x} - 2x - 1) + 4x$$

$$2e^{2x} - 2 \stackrel{?}{=} 2e^{2x} - 4x - 2 + 4x$$

Since the functions on the left and right are the same function, $y = e^{2x} - 2x - 1$ **is** a solution to the differential equation.

Differential equation:

$$x \frac{dy}{dx} = 7xy + y$$

Interpretation:

There is a function $y(x)$ that makes the left-hand side and the right-hand side into the same function.

To check whether a given function satisfies the differential equation, plug it in for everything with a “ y ”: y itself and $\frac{dy}{dx}$.

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Is $y = xe^{7x+9}$ a solution to the differential equation?

Differential equation: $x \frac{dy}{dx} = 7xy + y$

Function: $y = xe^{7x+9}$

Differential equation: $x \frac{dy}{dx} = 7xy + y$

Function: $y = xe^{7x+9}$

If $y = xe^{7x+9}$, then $\frac{dy}{dx} = x(7e^{7x+9}) + e^{7x+9} = (7x+1)e^{7x+9}$. We replace all terms depending on y in the differential equation with these functions.

$$x \frac{dy}{dx} = 7xy + y$$

$$x(7x+1)e^{7x+9} = 7x(xe^{7x+9}) + xe^{7x+9}$$

$$x(7x+1)e^{7x+9} = x(7x+1)e^{7x+9}$$

The left and right hand side of the equation give the same function, so our function $y = xe^{7x+9}$ satisfies the differential equation.

Which of the following solve the differential equation $\frac{dy}{dx} = \frac{x}{y}$?

A. $y = -x$

B. $y = x + 5$

C. $y = \sqrt{x^2 + 5}$

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A. $y = -x$

B. $y = x + 5$

C. $y = \sqrt{x^2 + 5}$

- ▶ If $y = -x$, then $\frac{dy}{dx} = -1$. Plugging into the differential equation yields: $-1 \stackrel{?}{=} \frac{x}{-x}$. Since the left and right are the same function (except for the single point when $x = 0$), we say $y = -x$ **solves** the differential equation.
- ▶ If $y = x + 5$, then $\frac{dy}{dx} = 1$. Plugging into the differential equation yields: $1 \stackrel{?}{=} \frac{x}{x+5}$. Since the left and right are **not** the same function, $y = x + 5$ **does not solve** the differential equation.
- ▶ If $y = \sqrt{x^2 + 5}$, then $\frac{dy}{dx} = \frac{2x}{2\sqrt{x^2 + 5}} = \frac{x}{\sqrt{x^2 + 5}}$. Plugging into the differential equation yields: $\frac{x}{\sqrt{x^2 + 5}} \stackrel{?}{=} \frac{x}{\sqrt{x^2 + 5}}$. Since the left and right are the same function, we say $y = \sqrt{x^2 + 5}$ **solves** the differential equation.

FIRST EXAMPLE OF A SEPARABLE DE

Definition

A separable differential equation is an equation for a function $y(x)$ that can be written in the form

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

(It may take some rearranging to get the equation into this form.)

For example:

$$y^2 \cdot \frac{dy}{dx} = 4x$$

FIRST EXAMPLE OF A SEPARABLE DE

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$$\begin{aligned}y^2 \cdot \frac{dy}{dx} &= 4x \\ \int \left(y^2 \cdot \frac{dy}{dx} \right) dx &= \int 4x \, dx \\ \int y^2 \, dy &= 2x^2 + C \\ \frac{1}{3}y^3 &= 2x^2 + C \\ y^3 &= 6x^2 + 3C \\ y(x) &= \sqrt[3]{6x^2 + 3C} \\ y(x) &= \sqrt[3]{6x^2 + D}\end{aligned}$$

Here C and D are arbitrary constants.

GENERAL METHOD FOR SOLVING SEPARABLE DES

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$$\int \left(g(y(x)) \cdot \frac{dy}{dx} \right) dx = \int f(x) dx$$

y -substitution:

$$\int g(y) dy = \int f(x) dx$$

GENERAL METHOD FOR SOLVING SEPARABLE DES

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

Shorthand:

$$g(y) \cdot \frac{dy}{dx} = f(x)$$

$$g(y) \, dy = f(x) \, dx$$

$$\int g(y) \, dy = \int f(x) \, dx$$

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3. Solve explicitly for y .

$$y = \frac{1}{-\frac{1}{2}x^2 - C}$$

$$\frac{dy}{dx} = (xy)^4, \quad y(0) = \frac{1}{2}$$

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$$\frac{dy}{dx} = x^4 y^4$$

$$y^{-4} dy = x^4 dx$$

$$\int y^{-4} dy = \int x^4 dx$$

$$\frac{1}{-3} y^{-3} = \frac{1}{5} x^5 + C$$

$$\frac{1}{y^3} = -3 \left(\frac{1}{5} x^5 + C \right)$$

$$y = \frac{1}{-\sqrt[3]{3 \left(\frac{1}{5} x^5 + C \right)}}$$

$$y(0) = -\sqrt[3]{\frac{1}{3 \left(\frac{1}{5} x^5 + C \right)}} \Big|_{x=0}$$

$$\frac{1}{2} = -\sqrt[3]{\frac{1}{3C}}$$

$$2 = -\sqrt[3]{3C}$$

$$3C = -8$$

$$y(x) = -\sqrt[3]{\frac{1}{\frac{3}{5} x^5 - 8}}$$

$$= \sqrt[3]{\frac{1}{8 - \frac{3}{5} x^5}}$$

$$\frac{dy}{dx} = y(4x^3 - 1) \quad y(0) = -2$$



$$\frac{dy}{dx} = y(4x^3 - 1) \quad y(0) = -2$$

$$\frac{1}{y} dy = (4x^3 - 1) dx$$

$$\int \frac{1}{y} dy = \int (4x^3 - 1) dx$$

$$\log |y| = x^4 - x + C$$

$$\text{When } x = 0, \log |-2| = 0^4 - 0 + C$$

$$C = \log 2$$

$$|y(x)| = e^{x^4 - x + \log 2}$$

$$y(x) = e^{x^4 - x + \log 2} \quad \text{or} \quad y(x) = -e^{x^4 - x + \log 2}$$

$$y(x) = -e^{x^4 - x + \log 2} = -2e^{x^4 - x} \quad \text{to make } y(0) = -2$$

Let a and b be any two constants. We'll now solve the family of differential equations

$$\frac{dy}{dx} = a(y - b)$$

using our mnemonic device.

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$$\frac{dy}{y - b} = a \, dx$$

$$\int \frac{dy}{y - b} = \int a \, dx$$

$$\log |y - b| = ax + c$$

$$|y - b| = e^{ax+c} = e^c e^{ax}$$

$$y - b = \pm e^c e^{ax} = C e^{ax}$$

where the constant C can be any real number. (Even $C = 0$ works, i.e. $y(x) = b$ solves $\frac{dy}{dx} = a(y - b)$.) Note that when $y(x) = C e^{ax} + b$ we have $y(0) = C + b$. So $C = y(0) - b$ and the general solution is

$$y(x) = \{y(0) - b\} e^{ax} + b$$

Linear First-Order Differential Equations

Let a and b be constants. The differentiable function $y(x)$ obeys the differential equation

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if and only if

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Find a function $y(x)$ with $y' = 3y + 7$ and $y(2) = 5$.

To avoid re-inventing the wheel, we'll use our equation. But first, we should re-write our differential equation so the formatting matches.

$$\begin{aligned}\frac{dy}{dx} &= 3 \left(y + \frac{7}{3} \right) \\ a &= 3, \quad b = -\frac{7}{3} \\ y(x) &= Ce^{3x} - \frac{7}{3}\end{aligned}$$

Since we aren't given $y(0)$, we can't use the theorem as a shortcut to find C . We'll do it the old-fashioned way.

$$\begin{aligned}5 &= y(2) = Ce^{3(2)} - \frac{7}{3} \\ \frac{22}{3} &= Ce^6 \\ C &= \frac{22}{3e^6} \\ y(x) &= \frac{22}{3e^6} e^{3x} - \frac{7}{3}\end{aligned}$$

The rate at which a medicine is metabolized (broken down) in the body depends on how much of it is in the bloodstream. Suppose a certain medicine is metabolized at a rate of $\frac{1}{10}A \mu\text{g/hr}$, where A is the amount of medicine in the patient. The medicine is being administered to the patient at a constant rate of $2 \mu\text{g/hr}$. If the patient starts with no medicine in their blood at $t = 0$, give the formula for the amount of medicine in the patient at time t . What happens to the amount over time?

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The rate of change of the amount of medicine in the patient is given by how quickly the medicine is being administered, minus how quickly it is metabolized:

$$\frac{dA}{dt} = 2 - \frac{1}{10}A$$

Linear First-Order Differential Equations

Let a and b be constants. The differentiable function $y(x)$ obeys the differential equation

$$\frac{dy}{dx} = a(y - b)$$

if and only if

$$y(x) = \{y(0) - b\} e^{ax} + b$$

$$\frac{dA}{dt} = 2 - \frac{1}{10}A = -\frac{1}{10}(A - 20) \quad A(0) = 0$$

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if and only if

$$y(x) = \{y(0) - b\} e^{ax} + b$$

$$\frac{dA}{dt} = 2 - \frac{1}{10}A = -\frac{1}{10}(A - 20) \quad A(0) = 0$$

$$a = -\frac{1}{10}, \quad b = 20$$

$$A(t) = (A(0) - 20)e^{-t/10} + 20$$

$$A(t) = -20e^{-t/10} + 20$$

This is an increasing function, with $\lim_{t \rightarrow \infty} A(t) = 20$. So the amount of medicine initially increases, but eventually almost holds steady at 20 μg .

Included Work

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